

Exponentiation II

Time limit: 1.00 s

Memory limit: 512 MB

Your task is to efficiently calculate values a^{b^c} modulo $10^9 + 7$.

Note that in this task we assume that $0^0 = 1$.

Input

The first input line has an integer n : the number of calculations.

After this, there are n lines, each containing three integers a , b and c .

Output

Print each value a^{b^c} modulo $10^9 + 7$.

Constraints

- $1 \leq n \leq 10^5$
- $0 \leq a, b, c \leq 10^9$

Example

Input	Output
3 3 7 1 15 2 2 3 4 5	2187 50625 763327764